

Simran Desai

UX Engineer · Design Engineer · Full-Stack Software Engineer

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UX engineering portfolio

Seattle, WA

SUMMARY

Full-stack engineer with 3+ years of experience designing and building AI-powered platforms and enterprise web applications for global data-center supply chain and capacity planning. I work across workflow research, interaction design, coded prototyping, frontend systems, agentic experiences, cloud infrastructure, UAT, and production iteration.

EXPERIENCE

Software Engineer I & II — Microsoft · Redmond, WA

September 2022 – Present

- Designed and deployed an AI-powered order explicability workflow using Azure AI Foundry and Angular, replacing fragmented email triage with notifications, blocker visualization, reservation history, deeplinks, and guarded remediation. The broader program targeted an 8% improvement in on-time reservations; that outcome is not attributed solely to the interface.
- Architected a React and Cosmos DB plan-explicability experience that reduced a supported capacity-planning analysis cycle from 160 minutes to 10 minutes and carried an estimated savings of approximately 20 hours per month; presented results in a senior leadership business review.
- Built a C# Azure AI Foundry agent for plan-audit history with agent-to-agent integration, PostgreSQL prefiltering, and dynamic query generation, reducing audit investigation time by 80% and distributing the implementation as a shared library.
- Engineered a DSPy Python agent with a feedback loop on a Kubernetes-hosted AI platform to generate advanced SQL drill-down queries, doubling the analytical capability available to SQL-based report templates.
- Redesigned the data-first loading architecture for supply-chain analytics, reducing load time by 82% from 45 seconds to 8 seconds, growing active users by 30%, and reducing new-user ramp-up time by 50%.
- Built reusable React controls, visualizations, grids, and event contracts as the first production use case on an internal AI canvas, enabling faster onboarding for later teams and supporting a 4–5× increase in daily tool adoption through end-user UAT.
- Automated Azure resource provisioning using Bicep infrastructure as code, deployment slots, Cosmos DB feature flags, and Azure Front Door routing, enabling zero-downtime deployments and reducing rollback time by 40 minutes.

Software Engineering Intern — Microsoft · Redmond, WA

May 2021 – August 2021

- Designed and launched an issue-management system that reduced deployment issue identification and mitigation time by 95% for internal stakeholders.
- Gathered requirements from automation teams and delivered an MVP within 12 weeks, increasing internal satisfaction and reducing manual work.

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EXPERIENCE, CONTINUED

Explore Intern — Microsoft · Redmond, WA	May 2020 – August 2020
- Built reusable UI components that improved accessibility compliance and reduced development time by 99%, enabling faster delivery of internal tools. - Defined KPIs and created a customer-success survey; applied the feedback to improve supplier interaction workflows through targeted UX changes.	

EDUCATION

Bachelor of Science in Computer Science	2022
University of Illinois Urbana-Champaign	
GPA: 3.77 / 4.00 · Dean’s List · RTC Fellow	

CERTIFICATIONS & AWARDS

Lean Six Sigma White Belt	2024
Microsoft Global Hackathon · 3rd Place	2021

SKILLS

UX engineering: Workflow research, interaction design, coded prototyping, agentic UI/UX, information architecture, design systems, accessibility, UAT, telemetry-driven iteration
Languages and frameworks: C#, Python, TypeScript, SQL, Angular, React, DSPy, HTML, CSS
AI and agentic systems: Azure AI Foundry, agent-to-agent integration, dynamic query generation, feedback-loop design, DSPy
Cloud and infrastructure: Azure App Service, Azure Front Door, Azure Application Insights, Bicep, Kubernetes, Cosmos DB, PostgreSQL
DevOps and delivery: Infrastructure as code, deployment slots, feature flags, zero-downtime deployment, CI/CD, cron automation, Git, Azure DevOps
Data and analytics: Power BI, SQL drill-down analysis, P90 latency monitoring, product telemetry

Public portfolio résumé. Personal phone number omitted intentionally. Program-level outcomes are distinguished from impact attributable solely to an interface.